

Pretty-print ☐

```

{
  "_meta": {
    "version": "odin_v1251_champion",
    "codename": "CORNERSTONE",
    "description": "Absolute champion config – best WF Brier (0.08880) across 267 honing iterations (v1083-v1349). Walk-forward validated on ODIN_MODEL_READY_v1070_T1_2015on_ENRICHED.csv.",
    "selection_basis": "Scanned all 267 _honed_ensemble.json and _honed.json configs. v1250+ series has TRUE walk-forward validation (v1083-v1249 had WF=VAL bug). v1251 pure ODIN beats ensemble on WF metrics.",
    "lineage": "v10.66 production → v1083-v1249 honing chain → v1249 anchor → v1250-v1251 (engine v5.2.0, seed 10001)",
    "champion_metrics": {
      "walk_forward_brier": 0.08880,
      "walk_forward_auc": 0.9082,
      "val_brier": 0.10022,
      "val_auc": 0.9190,
      "trainable_weights": 44,
      "total_parameters": 54,
      "engine_version": "5.2.0",
      "honoring_seed": 10001,
      "training_data": "ODIN_MODEL_READY_v1070_T1_2015on_ENRICHED.csv",
      "timestamp": "2026-02-21T17:36:24"
    },
    "comparison_table": {
      "v1251_honed_WF_brier_0.08880": "CHAMPION – best WF Brier + AUC across all configs",
      "v1250_honed_WF_brier_0.08903": "Runner-up, marginally worse WF",
      "v1251_ensemble_WF_brier_0.08886": "Ensemble adds HINT blend, slightly worse WF",
      "v1349_val_brier_0.10051_AUC_0.9002": "Overhoned – 100 iterations degraded AUC from 0.919→0.900",
      "v10.66_test_brier_0.0948": "Legacy production baseline (33 params, no WF validation)"
    },
    "known_concerns": {
      "class1_resubmission_boost": "Negative (-1.38) – semantically should be positive for Class 1 CRL. WF-validated but domain-questionable. Inherited from v1249 anchor chain.",
      "indication_onc_boost": "Negative (-0.93) – compensated by btd_oncology_boost (+2.11). Net oncology+BTd = +1.18 which is directionally correct.",
      "ta_low_risk_boost": "Negative (-1.87) – semantically wrong. May be compensating for other TA features. Monitor.",
      "priority_review_weight": "2.53 vs v10.66 production 0.55 (4.6x). Consistently honed high across all seeds. May reflect real PR signal or overfit to PR-correlated events.",
      "manufacturing_risk_penalty": "Effectively zero (-1e-06). Absorbed by interaction terms ix_prior_crl_x_mfg_risk and ix_inexperienced_x_mfg_risk.",
      "adcom_low_penalty": "Effectively zero (-1e-06). Surprising – negative AdCom votes historically very predictive."
    },
    "deployment_notes": "Use this as the MCP ODIN scorer weight file. The isotonic_calibration (Platt scaling) should be applied as the final step: p_calibrated = 1/(1+exp(-(a*logit + b))). The HINT ensemble blend (odin_weight, hint_weight) requires historical CRL rate data."
  },
  "base_logit": 1.6877965449124204,
  "penalties": {
    "snda_base_penalty": -3.0,
    "snda_pediatric_base_penalty": -0.06800629686518996,
    "prior_crl_penalty": -3.8551249937004815,
    "inexperienced_sponsor_penalty": -0.9877390491325875,
    "manufacturing_risk_penalty": -1e-06,
    "form_483_penalty": -1.0032841165652016,
    "ema_cmc_flag_penalty": -1.7637221258168696,
    "cmc_extension_penalty": -1.2804336072183253,
    "adcom_mid_penalty": -0.4388235981850125,
    "adcom_low_penalty": -1e-06,
    "s22_pediatric_pk_penalty": -1.742907536222042,

```

Pretty-print

```
"novice_sponsor_high_risk_ta_penalty": -0.30000001192092896,
"gene_therapy_penalty": -0.8653127123308743,
"single_arm_study_penalty": -3.5955999549937543,
"surrogate_endpoint_penalty": -0.2574345660486174,
"prior_crl_count_penalty": -0.7014723992715777,
"safety_severity_penalty": -0.15358569077830592,
"ppm_penalty": -3.799999952316284,
"eu_approved_2026_penalty": -0.17461282956209842,
"psychedelics_penalty": -1e-06,
"hoeg_era_constant": -0.10000000149011612,
"accel_approval_2025plus_penalty": -1.045293387445537,
"experienced_sponsor_2026_reduction": -1.389068962125728,
"double_crl_penalty": -1e-06,
"class1_resubmission_boost": -1.3777807269584754
},

"boosts": {
  "btd_weight": 0.20790020747697294,
  "orphan_weight": 0.03999999910593033,
  "priority_review_weight": 2.5256676705876178,
  "fast_track_weight": 1.1936271681297903,
  "accelerated_approval_weight": 0.3895689176206382,
  "experienced_sponsor_boost": 1.2455899836706399,
  "adcom_high_boost": 3.2669080748307007,
  "eu_approved_boost": 0.2916146425428995,
  "btd_oncology_boost": 2.1105462377523563,
  "ta_very_high_boost": 1.2788227468655247
},

"ta_weights": {
  "ta_adjustment_weight": -0.05641338337930061,
  "ta_high_risk_penalty": -0.9626554744429417,
  "ta_mod_risk_penalty": -0.5079140606058509,
  "ta_low_risk_boost": -1.86914877334142,
  "indication_onc_boost": -0.9253040106386666
},

"signal_weights": {
  "s23_insider_weight": 0.5861676961239876,
  "s6_hiring_weight": 0.9371508494347254,
  "social_weight": 0.12445292250731003
},

"hint_ensemble": {
  "odin_weight": 0.7520477981855773,
  "hint_weight": 0.6684420832053102,
  "hint_crl_rate_penalty": -0.2005418595851598
},

"interaction_terms": {
  "ix_prior_crl_x_mfg_risk": -0.25764701477541224,
  "ix_prior_crl_x_form483": -0.48206294901375923,
  "ix_gene_therapy_x_mfg_risk": -0.15000000596046448,
  "ix_inexperienced_x_mfg_risk": -3.0,
  "ix_single_arm_x_surrogate": 2.801338275288041,
  "ix_btd_x_single_arm": 1.7816286536277133
},

"isotonic_calibration": {
  "type": "platt",
  "a": -0.026989505605139973,
  "b": 0.5827207756734252
},
```

Pretty-print

```
"TIER_1": {"min_prob": 0.85, "action": "LONG", "sizing": "FULL", "exit": "1-5"},
"TIER_2": {"min_prob": 0.65, "action": "CAUTIOUS LONG", "sizing": "HALF", "exit": "T-7"},
"TIER_3": {"min_prob": 0.40, "action": "MONITOR", "sizing": "QUARTER", "exit": "T-7"},
"TIER_4": {"min_prob": 0.00, "action": "NO TRADE", "sizing": "ZERO", "exit": "N/A"}
},

"avoid_signals": [
  "EMA_CMC_FLAG",
  "HIRING_VOID_NDA",
  "PEDIATRIC_NO_PK",
  "CMC_EXTENSION",
  "INSIDER_CRITICAL",
  "PPM_FLAG",
  "GENE_THERAPY_CMC"
],

"flat_weights_for_scorer": {
  "base_logit": 1.6877965449124204,
  "snda_base_penalty": -3.0,
  "snda_pediatric_base_penalty": -0.06800629686518996,
  "prior_crl_penalty": -3.8551249937004815,
  "inexperienced_sponsor_penalty": -0.9877390491325875,
  "manufacturing_risk_penalty": -1e-06,
  "form_483_penalty": -1.0032841165652016,
  "ema_cmc_flag_penalty": -1.7637221258168696,
  "cmc_extension_penalty": -1.2804336072183253,
  "adcom_mid_penalty": -0.4388235981850125,
  "adcom_low_penalty": -1e-06,
  "s22_pediatric_pk_penalty": -1.742907536222042,
  "btd_weight": 0.20790020747697294,
  "orphan_weight": 0.03999999910593033,
  "priority_review_weight": 2.5256676705876178,
  "fast_track_weight": 1.1936271681297903,
  "accelerated_approval_weight": 0.3895689176206382,
  "class1_resubmission_boost": -1.3777807269584754,
  "experienced_sponsor_boost": 1.2455899836706399,
  "adcom_high_boost": 3.2669080748307007,
  "ta_adjustment_weight": -0.05641338337930061,
  "s23_insider_weight": 0.5861676961239876,
  "s6_hiring_weight": 0.9371508494347254,
  "social_weight": 0.12445292250731003,
  "odin_weight": 0.7520477981855773,
  "hint_weight": 0.6684420832053102,
  "hint_crl_rate_penalty": -0.2005418595851598,
  "ta_high_risk_penalty": -0.9626554744429417,
  "ta_mod_risk_penalty": -0.5079140606058509,
  "ta_low_risk_boost": -1.86914877334142,
  "indication_pain_penalty": -0.10000000149011612,
  "indication onc_boost": -0.9253040106386666,
  "novice_sponsor_high_risk_ta_penalty": -0.30000001192092896,
  "gene_therapy_penalty": -0.8653127123308743,
  "single_arm_study_penalty": -3.5955999549937543,
  "surrogate_endpoint_penalty": -0.2574345660486174,
  "prior_crl_count_penalty": -0.7014723992715777,
  "safety_severity_penalty": -0.15358569077830592,
  "ppm_penalty": -3.799999952316284,
  "eu_approved_boost": 0.2916146425428995,
  "eu_approved_2026_penalty": -0.17461282956209842,
  "psychedelics_penalty": -1e-06,
  "hoeg_era_constant": -0.10000000149011612,
  "accel_approval_2025plus_penalty": -1.045293387445537,
  "experienced_sponsor_2026_reduction": -1.389068962125728,
  "btd_oncology_boost": 2.1105462377523563,
  "ta_very_high_boost": 1.2788227468655247,
```

Pretty-print

```
"ix_prior_crl_x_mfg_risk": -0.25764701477541224,  
"ix_prior_crl_x_form483": -0.48206294901375923,  
"ix_gene_therapy_x_mfg_risk": -0.15000000596046448,  
"ix_inexperienced_x_mfg_risk": -3.0,  
"ix_single_arm_x_surrogate": 2.801338275288041,  
"ix_btd_x_single_arm": 1.7816286536277133  
}  
}
```